

Comparing Proportions

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This page reviews tests for comparing groups on a categorical outcome: the chi-square test and Fisher's exact test for contingency tables. It uses the chi-square reference distribution defined on the [Statistical Inference](#) page. This page is adapted from Vittinghoff et al. (2012), Chapter 3.

The HERS data

The examples on this page use the HERS data, which the [Comparing Means](#) page describes. The `rmb` R package includes the dataset; `haven::as_factor()` converts its Stata value labels to factors:

```
hers <- rmb::hers |> haven::as_factor()
hers |> head()
#> # A tibble: 6 x 37
#>   HT      age raceth nonwhite smoking drinkany exercise physact globrat poorfair
#>   <fct> <dbl> <fct>   <fct>   <fct>   <fct>   <fct>   <fct>   <fct>   <fct>
#> 1 plac~    70 Afric~ yes      no      no      no      much m~ good    no
#> 2 plac~    62 Afric~ yes      no      no      no      much l~ good    no
#> 3 horm~    69 White  no      no      no      no      about ~ good    no
#> 4 plac~    64 White  no      yes     yes     no      much l~ good    no
#> 5 plac~    65 White  no      no      no      no      somewh~ good    no
#> 6 horm~    68 Afric~ yes      no      yes     no      about ~ good    no
#> # i 27 more variables: medcond <dbl>, htnmeds <fct>, statins <fct>,
#> #   diabetes <fct>, dmpills <fct>, insulin <fct>, weight <dbl>, BMI <dbl>,
#> #   waist <dbl>, WHR <dbl>, glucose <dbl>, weight1 <dbl>, BMI1 <dbl>,
#> #   waist1 <dbl>, WHR1 <dbl>, glucose1 <dbl>, tchol <dbl>, LDL <dbl>,
#> #   HDL <dbl>, TG <dbl>, tchol1 <dbl>, LDL1 <dbl>, HDL1 <dbl>, TG1 <dbl>,
#> #   SBP <dbl>, DBP <dbl>, age10 <dbl>
```

Comparing two groups: categorical outcomes

Contingency tables

Example 0.1 (Exercise by treatment group in HERS). Table 1 cross-tabulates regular exercise at baseline by treatment group, as a [contingency table](#) with row percentages.

Table 1: Exercise by treatment group in HERS

```
hers |>
  gtsummary::tbl_cross(
    row = exercise,
    col = HT,
    label = list(
      exercise ~ "Exercises regularly",
      HT ~ "Treatment group"
    ),
    percent = "row"
  )
```

	Treatment group		Total
	placebo	hormone therapy	
Exercises regularly			
no	853 (50%)	842 (50%)	1,695 (100%)
yes	530 (50%)	538 (50%)	1,068 (100%)
Total	1,383 (50%)	1,380 (50%)	2,763 (100%)

The chi-square test

Definition 0.1 (Expected count under independence). Let a contingency table have r rows and c columns, with observed count O_{ij} in row i and column j , row totals R_i , column totals C_j , and grand total n . The **expected count** in cell (i, j) under independence is

$$E_{ij} \stackrel{\text{def}}{=} \frac{R_i C_j}{n}.$$

Example 0.2 (Expected count in a 2 x 2 table). In a table with $n = 100$, first-row total $R_1 = 30$, and first-column total $C_1 = 40$, the expected count in cell $(1, 1)$ is $E_{11} = 30 \cdot 40 / 100 = 12$.

Definition 0.2 (Pearson's chi-square test of independence). With observed counts O_{ij} and **expected counts** E_{ij} in an $r \times c$ contingency table, **Pearson's chi-square test** of the null hypothesis that the row and column variables are independent uses the statistic

$$X^2 \stackrel{\text{def}}{=} \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Its p-value is $\Pr(W \geq X^2)$, where W has the $\chi^2_{(r-1)(c-1)}$ distribution (**chi-square distribution**).

Theorem 0.1 (Large-sample null distribution of the chi-square statistic). *Let n observations be sampled independently and classified by two categorical variables, and let the two variables be independent. Then as $n \rightarrow \infty$, the distribution of X^2 (Definition 0.2) converges to the $\chi^2_{(r-1)(c-1)}$ distribution (Hogg et al. 2019, sec. 9.2, p. 440).*

Remark 0.1 (Small expected counts). The chi-square approximation is poor when some expected counts are small. A common rule of thumb asks for every E_{ij} to be at least 5.

Example 0.3 (Chi-square test of exercise by treatment group in HERS). The expected counts and statistic of Definition 0.2 for Table 1:

```
observed <- table(hers$exercise, hers$HT)
expected <- outer(rowSums(observed), colSums(observed)) / sum(observed)
x_squared <- sum((observed - expected)^2 / expected)
expected
#>      placebo hormone therapy
#> no    848.42      846.58
#> yes   534.58      533.42
c(
  x_squared = x_squared,
  p_value = pchisq(x_squared, df = 1, lower.tail = FALSE)
)
#> x_squared  p_value
#> 0.128054  0.720458
```

Every expected count is large, so the χ^2_1 approximation of Theorem 0.1 is reasonable.

`chisq.test()` with `correct = FALSE` reports the same values:

```
chisq.test(hers$exercise, hers$HT, correct = FALSE)
#>
#> Pearson's Chi-squared test
#>
#> data:  hers$exercise and hers$HT
#> X-squared = 0.1281, df = 1, p-value = 0.72
```

The p-value is large: the data give no evidence that exercise depends on treatment group, as randomization would lead us to expect.

Remark 0.2 (Yates' continuity correction). For a 2×2 table, `chisq.test()` applies Yates' continuity correction by default, which subtracts 0.5 from each $|O_{ij} - E_{ij}|$ before squaring, and so gives a smaller statistic than Definition 0.2. `correct = FALSE` turns the correction off.

Fisher's exact test

Definition 0.3 (Fisher's exact test). Take a 2×2 contingency table with cells a , b , c , and d as in [the contingency table definition](#), and hold its row and column totals fixed. Under independence of the row and column variables, the probability that the top-left cell equals x is the hypergeometric probability

$$p(x) \stackrel{\text{def}}{=} \frac{\binom{a+b}{x} \binom{c+d}{a+c-x}}{\binom{n}{a+c}}.$$

Fisher's exact test of independence has p-value

$$\sum_{x: p(x) \leq p(a)} p(x),$$

the total probability of the tables with the same totals that are no more probable than the observed table.

Remark 0.3 (Why Fisher's exact test is exact). The p-value is exact: it comes from the null distribution itself, not from a large-sample approximation, so the test is valid even when expected counts are small, and it is often used for 2×2 tables in which some expected count is below 5 (Theorem 0.1). Other two-sided versions exist; this one is the version that R's `fisher.test()` computes.

Example 0.4 (Fisher’s exact test of exercise by treatment group in HERS). The p-value of Definition 0.3 for Table 1, computed from the hypergeometric probabilities with `dhyper()`:

```
a <- observed[1, 1]
row1 <- sum(observed[1, ])
row2 <- sum(observed[2, ])
col1 <- sum(observed[, 1])
x <- max(0, col1 - row2):min(row1, col1)
p_x <- dhyper(x, m = row1, n = row2, k = col1)
p_a <- dhyper(a, m = row1, n = row2, k = col1)
sum(p_x[p_x <= p_a * (1 + 1e-7)])
#> [1] 0.72528
```

The tolerance `1e-7` keeps tables whose probability equals $p(a)$ up to rounding error, as `fisher.test()` does. `fisher.test()` reports the same p-value:

```
fisher.test(hers$exercise, hers$HT)
#>
#> Fisher's Exact Test for Count Data
#>
#> data: hers$exercise and hers$HT
#> p-value = 0.725
#> alternative hypothesis: true odds ratio is not equal to 1
#> 95 percent confidence interval:
#> 0.879664 1.202192
#> sample estimates:
#> odds ratio
#> 1.02836
```

With counts this large, the exact p-value is close to the chi-square p-value of Example 0.3.

Measures of association for 2×2 tables

Tests of independence say whether two binary variables are associated, but not how strongly. Risk differences, risk ratios, and odds ratios measure the strength of the association; see [Odds Ratios and Relative Risks](#).

References

- Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman. 2019. *Probability and Statistical Inference*. Tenth edition. Pearson.
- Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.